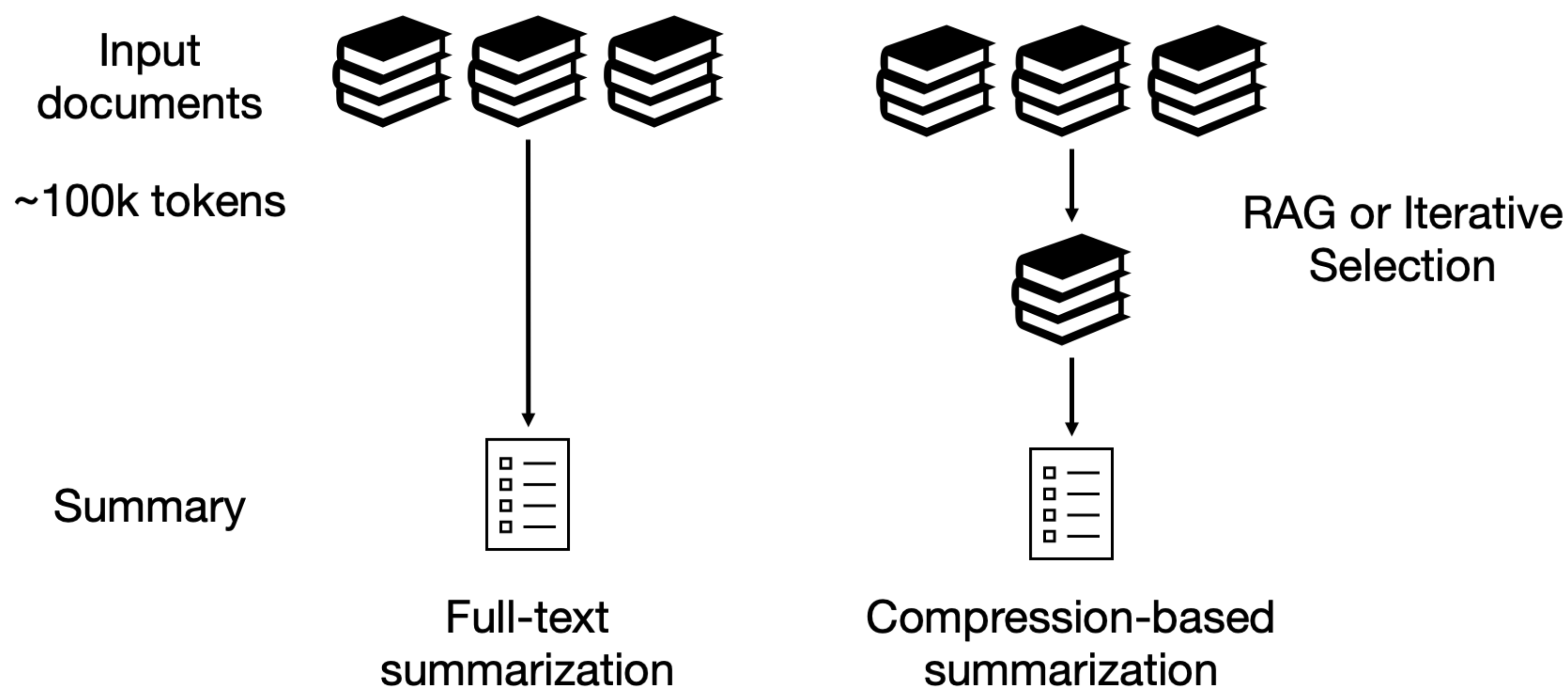


Scaling Multi-Document Event Summarization

Evaluating Compression vs. Full-Text Approaches



Full input (long context window) or compressed input (short context window)?



Compression

- Efficient
- Scalable
- Retrieval, Hierarchical, Incremental



Full-text

- Effective
- Utilizes long-context windows of recent LLMs

Base models

- Llama-3.1 (8B, 70B)
- Command-R (32B)
- Jamba 1.5 Mini (52B MoE, 12B active)

Datasets

- SummHay (Laban et al., 2024)
- Background (Pratapa et al., 2023)
- WCEP (Gholipour Ghalandari et al., 2020)

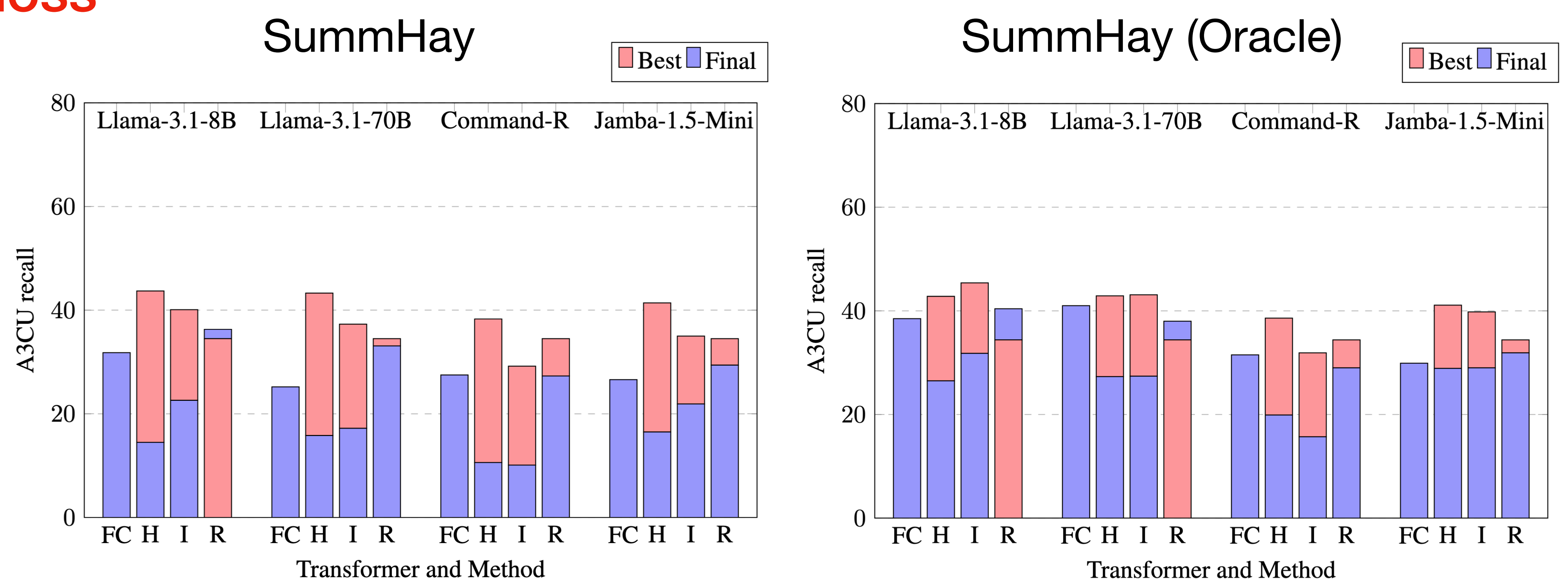
Results

	Llama-3.1-8B	Llama-3.1-70B	Command-R	Jamba-1.5-Mini
SummHay	-53% -32% +4% 33.9	-44% -37% +17% 31.1	-63% -63% 0% 30.4	-54% -32% +6% 32.9
SummHay (oracle)	-27% -17% +1% 37.1	-35% -29% -6% 41.8	-33% -47% -3% 32.6	-16% -18% +3% 35.1
Background	-16% -36% +8% 15.6	-9% -31% -3% 16.1	-10% -6% +15% 10.3	-17% -15% -11% 12.3
WCEP	-15% -22% -3% 30.7	-13% -22% -2% 31.1	-11% -13% -1% 28.9	-2% -13% -1% 29.3

Relative performances of **hierarchical**, **incremental** and **retrieval** compared to full-text baseline (AutoACU F1)

Tracking information loss

We measure A3CU recall scores of final system summary and the *best* intermediate output.



Proposals for future work

- Hybrid approaches that combine compression with long-context windows could be more efficient and effective.
- We need automatic metrics that do not rely on human-written references.

